

llmXive Automated Review of MobileGym: A Verifiable and Highly Parallel Simulation Platform for Mobile GUI Agent Research

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of citations supporting them.

1. Misattribution of RL Methodology (GRPO) In Section 5.1 (“Sim-to-Real Transfer”), the authors state: “Fine-tuned Qwen3-VL-4B-Instruct with GRPO~\cite{GRPO}...”. The bibliography entry for \cite{GRPO} points to “DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models” (Shao et al., 2024). This is a factual error. The GRPO (Group Relative Policy Optimization) algorithm was introduced in the DeepSeek-R1 technical report (2025) to improve reasoning capabilities. The “DeepSeekMath” paper (2024) primarily utilizes PPO (Proximal Policy Optimization) and does not introduce GRPO. Attributing the specific RL algorithm used to achieve the reported -12.8pt gain to a paper that does not contain it undermines the reproducibility and scientific accuracy of the training claim. The citation must be updated to the correct source (e.g., the DeepSeek-R1 report or the specific GRPO implementation paper).

2. Hallucinated or Future-Dated Model Citation In Section 5.1 and Appendix D (“Detailed VLM-Judge Misjudgment Audit”), the authors claim: “GPT-5.4~\cite{GPT54} also found 12/118 errors.” The bibliography lists \cite{GPT54} as “GPT-5.4 Thinking System Card” by OpenAI, dated March 2026. Given that the paper is a preprint (arXiv:2605.26114) and the current real-world date precedes the release of GPT-5, this citation refers to a model that does not exist in the public domain. If this is a “future-dated” paper (as the arXiv ID suggests), the claim that a specific, unreleased model was used for an audit is unverifiable and scientifically unsound. If the authors intended to use a known model (e.g., GPT-4o or GPT-4.5), the citation and text must be corrected to reflect a real, available model. Relying on a non-existent model for a key validation metric (VLM judge error rate) invalidates the claim of robustness.

3. Inconsistent Characterization of Baseline RL Capabilities Table 1 (“Comparison of mobile GUI agent benchmarks”) characterizes AndroidWorld as having “Limited” online RL readiness. However, the paper cites DigiRL (Bai et al., NeurIPS 2024), which explicitly demonstrates large-scale online RL training on AndroidWorld. The existence of DigiRL, which uses AndroidWorld for exactly the purpose the table claims is “Limited,” creates a contradiction. The claim that AndroidWorld is “Limited” for RL is factually weak given the evidence provided in the paper’s own bibliography. The table should be revised to accurately reflect that AndroidWorld is used for online RL (as shown by DigiRL), perhaps noting specific limitations (e.g., speed, state reset) rather than a blanket “Limited” status that contradicts the cited work.

4. Verification of “95.1% Retained Gain” The abstract and Section 5.1 claim “95.1% of the simulation-side training gain is retained.” This is calculated as $(\text{Real Gain} / \text{Sim Gain}) = 40.7 / 42.8 \approx 0.9509$. While the arithmetic is correct, the claim relies on the “59-task signal-bucket subset” (Sec 5.1). The paper states 8 tasks were dropped for irreproducibility. The claim of “95.1% retention” is a strong statistical assertion based on a filtered subset. While not a citation error, the text should explicitly clarify that this high retention rate applies *only* to the subset of tasks that were reproducible on real devices, as the dropped tasks

(189 stable-fail – 8 dropped) might skew the overall transferability picture. The current phrasing risks overgeneralizing the result to the entire benchmark.

Action items.

- [science] The citation for GRPO (line ~130, Sec 5.1) attributes the method to ‘DeepSeek-Math’ (Shao et al., 2024). However, GRPO (Group Relative Policy Optimization) was introduced in ‘DeepSeek-RL’ (2025) for reasoning tasks. The ‘DeepSeekMath’ paper primarily focuses on PPO and mathematical reasoning datasets. This citation likely misattributes the specific RL algorithm used for the 12.8pt gain.
- [writing] The claim that ‘GPT-5.4’ (cited as GPT54) was used to audit VLM judge errors (Sec 5.1, App D) is factually unsupported. The provided bibliography lists ‘GPT-5.4 Thinking System Card’ (OpenAI, March 2026). As of the current real-world date, GPT-5 has not been released. If this is a future-dated preprint, the citation must be verified against the actual model release or corrected to a known model (e.g., GPT-4o) to avoid hallucinated evidence.
- [writing] Table 1 claims AndroidWorld has ‘Limited’ online RL readiness, yet the paper cites ‘DigiRL’ (NeurIPS 2024) which explicitly uses AndroidWorld for online RL training. The characterization of AndroidWorld as ‘Limited’ for RL contradicts the existence of DigiRL’s methodology, which relies on the same environment for scalable training.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The provided image for Figure 1 is a generic mobile interface screenshot that fails to illustrate the specific action described in the caption, rendering the figure unusable for scientific verification.

Figure 2

The figure effectively illustrates the system’s interface with clear annotations, but the caption is marred by missing text placeholders (‘s’) that prevent it from clearly identifying the system being demonstrated.

Figure 5

The figure effectively illustrates the end-to-end workflow with clear visual steps, but the caption is severely flawed with missing system names and appears to be incorrectly copied from the Figure 4 caption, failing to accurately describe the specific content shown.

Figure 4

The figure effectively visualizes the system's composition model and state management workflow with clear diagrams and JSON examples. However, the caption contains a grammatical error where the system name is missing from the first sentence.

Figure 5

Figure 5 effectively illustrates the AnswerSheet protocol by contrasting the failure modes of free-text heuristics with the structured, deterministic verification of the AnswerSheet approach. The visual flow from GUI form filling to typed state and type-specific matching is clear, and the caption accurately describes the depicted workflow.

Figure 6

The bar chart effectively displays the success rate improvements, but the x-axis labels are truncated and the lack of error bars for the 4-seed averages limits the assessment of statistical significance.

Action items.

- [fatal] Figure 1: The rendered image is a screenshot of a mobile app's post creation interface (r/China_irl) and does not depict the 'grayed-out Post' button or the specific action described in the caption.
- [fatal] Figure 1: The image is a UI screenshot rather than a scientific figure; it lacks necessary annotations, arrows, or highlights to verify the claimed 'Step 13' action.
- [writing] Figure 2: The caption contains multiple instances of the placeholder 's' (e.g., 'Example screens from .', 'showing's configurable...') where the system name 'MobileGym' should appear, rendering the description grammatically incomplete.
- [writing] Figure 2: The caption text 'Example screens from .' is grammatically incomplete and lacks the specific name of the system or platform being illustrated.
- [writing] Figure 3: The caption contains multiple instances of missing text where the system name should appear (e.g., 'End-to-end workflow of .', 'Example screens from .', 'capabilities of .'). This makes the figure description grammatically incomplete and unclear.
- [writing] Figure 3: The caption text 'Figure 3: End-to-end workflow of .' appears to be a copy-paste error from Figure 4's caption ('System capabilities and state model of .'), as the image clearly depicts a workflow pipeline rather than a system architecture or state

model.

- [writing] Figure 4: The caption contains a missing noun in the first sentence (“System capabilities and state model of .”), likely due to a placeholder for the system name (MobileGym) that was not filled in.
- [writing] Figure 6: The x-axis labels (e.g., ‘Uplift’, ‘Mid’) are truncated and do not match the caption’s description of ‘Per-bucket Success Rate’, making it unclear which specific task buckets are being compared.
- [writing] Figure 6: The caption states ‘Sim columns are 4-seed averages’, but the figure lacks error bars to visualize the variance across these seeds, which is standard for reporting averages.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific shorthand and acronyms that are not consistently defined, creating barriers for readers outside the immediate niche of mobile GUI agent benchmarking.

In Section 5.3 (Evaluation Protocol), the authors introduce a suite of metrics: SR, PR, FC, USE, and OT. While standard in this specific sub-field, these are not defined at their first appearance. A generalist reader cannot infer that “USE” stands for “Unexpected Side Effects” or “OT” for “Overdue Termination” without guessing. These must be spelled out immediately upon introduction.

In Section 4.1, the term EFSM is used to describe the navigation model. This should be expanded to “Extended Finite State Machine” on first use. Similarly, in the Appendix (System Implementation Details), the text mentions HMR (Hot-Module Replacement) in the context of Vite. This is a specific build-tool concept that should be defined for readers who may not be frontend engineers.

The abbreviation OOD appears in Section 6.1 regarding generalization. While common in ML, it should be explicitly defined as “Out-of-Distribution” when first used in the main text to ensure clarity.

Finally, the unit pt is used frequently (e.g., “gains -12.8 pt”) to denote percentage points. While concise, spelling this out as “percentage points” in the Abstract and key results sections would improve accessibility for a broader scientific audience. The current density of undefined acronyms forces the reader to constantly flip between the text and the tables or appendices to decode the prose.

Action items.

- [writing] Define ‘SR’, ‘PR’, ‘FC’, ‘USE’, and ‘OT’ at first use in Section 5.3 (Evaluation

Protocol) instead of assuming reader familiarity with these specific benchmark metrics.

- [writing] Replace the acronym ‘EFSM’ in Section 4.1 with ‘Extended Finite State Machine’ on first occurrence to aid non-formal-methods readers.
- [writing] Define ‘HMR’ (Hot-Module Replacement) in the Appendix or Section 4.1, as it is a specific build-tool term that may be opaque to general AI researchers.
- [writing] Clarify ‘OOD’ in Section 6.1 (Sim-to-Real Transfer) as ‘Out-of-Distribution’ before using the abbreviation, as it is a critical concept for the transfer claim.
- [writing] Replace ‘pt’ with ‘percentage points’ in the Abstract and Section 6.1 to ensure clarity for readers less familiar with RL evaluation shorthand.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent architecture for a browser-hosted mobile simulation, but several key empirical claims suffer from logical gaps regarding dataset comparability and metric definitions.

First, the central claim of “95.1% retained gain” in the Sim-to-Real transfer (Abstract, Sec 4.2) is logically fragile. The authors calculate this by comparing a -40.7pt gain on a 59-task real-device subset to a -42.8pt gain on a simulation subset. However, the text does not explicitly confirm that the simulation baseline (33.9%) and post-training score (76.7%) used for this specific calculation were derived from the *exact same* 59-task subset. If the simulation numbers are averaged over the full 256-task test set (as implied by the general results in Table 1), the comparison is invalid because the difficulty distribution of the 59-task subset may differ significantly from the full set. Without explicit confirmation that the simulation metrics for the retention calculation are subset-specific, the 95.1% figure is an unsupported extrapolation.

Second, the “Unexpected Side Effects” (USE) metric (Sec 3.2, Table 1) lacks a clear logical mechanism for distinguishing between “side effects” and “valid but irrelevant state changes.” The system uses JSON state comparison to detect mutations. However, in a complex app, many state changes (e.g., cache updates, background sync flags) occur naturally during task execution. The paper claims to detect “mutations outside task goals,” but does not explain how the system defines the “task goal” boundary in the JSON schema to exclude these normal background operations. If the system flags any state change not explicitly in the “AnswerSheet” as a side effect, the metric may be measuring “state volatility” rather than “unintended side effects,” undermining the validity of the USE percentages reported in Table 1.

Finally, the difficulty stratification (L1-L4) is calibrated using 8 reference models (Sec

3.4). The logic assumes that the performance of these 8 models is a sufficient proxy for the “true” difficulty distribution. However, the paper notes in Appendix A.5 that shifting to 4 models changes the bucket counts significantly (e.g., L4 drops from 80 to 74 tasks). While the authors argue the separation remains, the sensitivity of the stratification to the choice of reference models suggests the difficulty labels are not robust. The conclusion that L4 “isolates the frontier” is therefore contingent on a specific, somewhat arbitrary choice of reference models, which weakens the generalizability of the benchmark’s difficulty claims. Action items.

- [science] The “95.1% retained gain” claim compares a -40.7pt real-device gain (on 59 tasks) to a -42.8pt sim gain. The text fails to explicitly state that the sim gain was calculated on the *same* 59-task subset. If the sim gain is from the full 256-task set, the comparison is logically invalid.
- [science] The “Unexpected Side Effects” (USE) metric detects JSON mutations outside task goals. The paper does not explain how the system distinguishes between unintended side effects and valid background state changes (e.g., cache updates) that are not part of the immediate task goal but are normal app behavior.
- [science] The L1-L4 difficulty stratification is sensitive to the choice of reference models (8 vs 4 models changes bucket counts significantly). The claim that L4 “isolates the frontier” relies on a specific, somewhat arbitrary model selection, weakening the robustness of the difficulty labels.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims regarding the fidelity and scalability of MobileGym that extend beyond the provided empirical evidence.

First, the central claim of “95.1% retained gain” in the Sim-to-Real transfer (Abstract, §4.2) is derived from a highly curated subset of 59 tasks. The authors explicitly state in Appendix §4.2 that 8 tasks were dropped due to irreproducibility and 189 tasks were “stable-fail” (where neither base nor trained models succeeded). By calculating the retention rate only on the “signal-bucket” (Uplift, Mid, Stable-pass) and ignoring the 189 tasks where the simulation failed to predict real-world failure, the paper overstates the generalizability of the transfer. The claim implies a high-fidelity transfer across the benchmark, whereas the data only supports high-fidelity transfer for a specific, pre-selected slice of tasks where the simulation was already “correct” about the outcome. The text should be revised to strictly limit this claim to the evaluated subset.

Second, the assertion that the platform enables “scalable online RL” (Abstract, §1) is

supported by a single experiment running 96 parallel instances. While the architecture (browser-based, JSON state) theoretically supports this, the paper presents the 96-instance run as evidence of “scalability” without demonstrating the system’s behavior at the claimed “hundreds of instances” scale or providing a wall-clock comparison of a full training epoch against the emulators it critiques. The claim of “scalability” is currently an architectural promise rather than an empirically validated result.

Finally, the framing of the Sim-to-Real results in §4.2 risks over-interpretation. While the *relative* gain is retained, the *absolute* performance gap between simulation (22.2%) and real device (72.9%) is substantial. The paper states “real-device execution retains 95.1% of the simulation-side training gain” but does not sufficiently emphasize that the base model performed significantly better on the real device (32.2%) than in simulation (9.4%). This suggests the simulation may be underestimating the model’s capability or that the real-device environment is easier for this specific task set, rather than the simulation being a perfect proxy. The conclusion that the simulation is “verifiable” and “highly parallel” is sound, but the specific quantitative claims about transfer fidelity require more nuanced qualification to avoid over-claiming the platform’s predictive power for the full benchmark. Action items.

- [writing] The paper makes several claims regarding the fidelity and scalability of MobileGym that extend beyond the provided empirical evidence. First, the central claim of “95.1% retained gain” in the Sim-to-Real transfer (Abstract, §4.2) is derived from a highly curated subset of 59 tasks. The authors explicitly state in Appendix §4.2 that 8 tasks were dropped due to irreproducibility and 189 tasks were “stable-fail” (where neither base nor trained models succeeded). By calculating the retention rate on

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents MobileGym, a simulation platform for mobile GUI agents, and includes a dedicated “Ethical Considerations” section. From a safety and ethics perspective, the work is generally sound in its intent to provide a “consequence-free” sandbox for testing. However, several areas require clarification to ensure responsible dissemination and usage.

First, the empirical results in the “High-Risk subset” (Table 5, Appendix) are concerning from a safety alignment perspective. The data indicates that frontier models (e.g., Gemini 3.1 Pro) can successfully execute payment and irreversible tasks with high success rates (64.3% and 71.4% respectively) without any mention of refusal or safety checks. While the simulation is sandboxed, the paper must explicitly address the dual-use risk: if agents are trained to maximize success on these tasks in simulation, they may learn to bypass safety guardrails when deployed on real devices. The authors should discuss whether

the training process (GRPO) inadvertently optimizes for “jailbreaking” behaviors and propose a mitigation strategy, such as incorporating safety rewards or refusal training in the benchmark.

Second, while the “Ethical Considerations” section mentions “digital-literacy education” as a positive use case, it lacks a concrete policy for preventing misuse. The platform enables the generation of synthetic data and the training of agents capable of navigating complex apps. There is a risk that this technology could be repurposed for automated social engineering, credential harvesting, or fraud. The authors should add a specific “Responsible Use” statement or a code of conduct for users of the MobileGym benchmark and the associated models, explicitly prohibiting the use of the platform for malicious automation.

Finally, the “Datasheet for Datasets” (§1) acknowledges biases, including an “English-speaking demographic focus” and “limited accessibility coverage.” The paper should elaborate on the safety implications of these biases. For instance, an agent trained primarily on English interfaces might fail catastrophically or behave unpredictably when interacting with non-English apps or accessibility features, potentially leading to user harm or data loss. A plan to expand the benchmark’s linguistic and accessibility coverage to mitigate these risks would strengthen the ethical standing of the work.

Overall, the paper is safe to publish provided these specific risks are acknowledged and mitigated through clearer discussion and usage guidelines.

Action items.

- [writing] The ‘High-Risk subset’ results (Table 5) show frontier models successfully executing payment tasks (64.3% SR) without explicit refusal mechanisms. The paper must explicitly discuss the dual-use risk of training agents to bypass safety guardrails in financial contexts and clarify if the ‘sandboxed’ nature of the simulation adequately mitigates the risk of transferring these behaviors to real-world deployment.
- [writing] Section ‘Ethical Considerations’ mentions ‘digital-literacy education’ as a use case but lacks a concrete plan for preventing the misuse of the generated synthetic data or the trained models for social engineering or automated fraud. A specific mitigation strategy or usage policy for the released benchmark and models is required.
- [writing] The ‘Datasheet for Datasets’ section (§1) notes an ‘English-speaking demographic focus’ and ‘limited accessibility coverage.’ The authors should elaborate on the potential for bias to cause safety failures for non-English speakers or users with disabilities when these agents are deployed in real-world scenarios, and propose a plan to address these gaps.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claim of high-fidelity Sim-to-Real transfer is promising but statistically underpowered due to sample size and selection constraints.

1. Selection Bias in Sim-to-Real Validation (§4.2): The authors validate the platform’s transferability using a “signal-bucket” subset of only 59 tasks (out of 256 test tasks), specifically chosen from Uplift, Mid, and Stable-pass categories. This non-random selection introduces a significant risk of selection bias. By excluding “Stable-fail” tasks (where the agent fails in both sim and real) and “Regression” tasks, the authors may be cherry-picking scenarios where the simulation dynamics align best with reality. The claim that “95.1% of the training gain is retained” is mathematically derived from this specific subset. To support this claim robustly, the authors must either: (a) report the transfer metrics on the full test set (or a large random sample), or (b) provide a rigorous statistical justification for why this specific subset is representative of the general distribution of task difficulties and failure modes.

2. Lack of Variance Reporting in Real-Device Experiments: While the simulation results in Table 1 include standard deviations (e.g., $58.8\% \pm 1.4$), the real-device evaluation results in §4.2 (e.g., “ $32.2\% \square 72.9\%$ ”) are presented as point estimates without confidence intervals or standard deviations. Given the small sample size (59 tasks) and the inherent stochasticity of real-device execution (network latency, OS background processes), the absence of variance metrics makes it impossible to assess the statistical significance of the observed gains. A t-test or bootstrap analysis is required to confirm that the observed transfer is not due to random chance.

3. Calibration of Difficulty Strata (§3.4): The L1-L4 difficulty stratification is a key contribution for benchmarking, yet the calibration process relies on 8 reference models. The paper lists the models in the appendix but does not explicitly detail the distribution of their performance or the specific thresholds used to define the strata boundaries in the main text. If the reference models are not diverse or if their performance is highly correlated, the resulting difficulty labels may be unstable. The authors should include a sensitivity analysis (beyond the 4-model check in the appendix) showing how the strata assignments change with different model sets, ensuring the benchmark’s difficulty labels are robust to the choice of reference agents.

4. Effect Size and Power: The improvement on the L4 (hardest) tasks is negligible (-0.9 pt). While this correctly identifies the frontier, the sample size for L4 is 80 tasks. The lack of improvement suggests the current RL approach hits a ceiling. However, without a power analysis, it is unclear if the study was designed to detect smaller effect sizes on these difficult tasks. The current evidence supports the claim that the platform *works* for easy/medium tasks but is inconclusive regarding its ability to drive improvements on the hardest tasks due to the flat performance curve.

Action items.

- [writing] The scientific evidence supporting the central claim of high-fidelity Sim-to-Real transfer is promising but statistically underpowered due to sample size and selec-

tion constraints. 1. Selection Bias in Sim-to-Real Validation (§4.2): The authors validate the platform’s transferability using a “signal-bucket” subset of only 59 tasks (out of 256 test tasks), specifically chosen from Uplift, Mid, and Stable-pass categories. This non-random selection introduces a significant risk of selection bias. B

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in the paper is generally sound in its presentation of point estimates and diagnostic metrics, but it lacks necessary rigor in variance reporting and significance testing for its key empirical claims.

First, regarding the benchmark results in Table 1 (Main Results), there is an inconsistency in the reporting of uncertainty. The proprietary model “Gemini 3.1 Pro” includes a standard deviation (± 1.4), while other models like “Doubao-Seed-2.0-Pro” and “Qwen3.6-Plus” do not. The footnote mentions that † marks single-run results, but it is unclear if the non-daggered open-source models were also single-run or if the variance was simply omitted. For a fair statistical comparison of model performance, especially when differences are small (e.g., 13.8% vs 15.4%), the absence of variance estimates for the majority of baselines weakens the statistical validity of the ranking. The authors should either report standard deviations for all models (via multiple seeds) or explicitly state which are single-run and treat the others as such in the discussion.

Second, the central claim of Sim-to-Real transfer (Section 4.2, Table 3 and Figure 3) relies on a comparison of gains: simulation gain of -42.8 pt versus real-device gain of -40.7 pt, resulting in a “95.1% retained gain.” This calculation is based on a subset of only 59 tasks (the “signal-bucket”). While the point estimates are close, the paper does not provide a statistical test (e.g., a paired t-test or a bootstrap confidence interval) to determine if the difference between the simulation gain and the real-device gain is statistically significant. Given the small sample size ($n=59$) and the inherent noise in real-device execution, claiming “95.1% retention” as a definitive metric without a confidence interval is statistically premature. The authors should provide a 95% confidence interval for the difference in gains to support the claim that the transfer is robust.

Finally, the difficulty stratification (L1-L4) described in Section 3.4 is calibrated using 8 reference models. While the appendix mentions a sensitivity check with 4 models, the main text does not list the specific 8 models used for the primary calibration. To ensure the reproducibility of the difficulty levels and the validity of the stratification, the full list of reference models and their individual performance distributions should be explicitly provided in the appendix or a supplementary table. This is crucial for verifying that the strata boundaries (e.g., $L1 \geq 75\%$) are not artifacts of a specific model selection bias.

Action items.

- [science] In Table 1 (Main Results), standard deviations are reported for some models (e.g., Gemini 3.1 Pro: ± 1.4) but not others (e.g., Doubao-Seed-2.0-Pro). Clarify if the latter are single-run results and justify the lack of variance reporting for the primary baselines to ensure statistical comparability.
- [science] The Sim-to-Real transfer claim (95.1% gain retention) relies on a small subset of 59 tasks. Report confidence intervals or a statistical test (e.g., paired t-test or bootstrap) for the difference between simulation gain (-42.8 pt) and real-device gain (-40.7 pt) to validate the ‘retention’ metric.
- [writing] The difficulty stratification (L1-L4) is calibrated using 8 reference models, but the specific models and their performance distributions are not fully detailed in the main text. Provide the full list of reference models and their mean SRs in the appendix to verify the robustness of the strata boundaries.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and technically dense narrative, effectively communicating the architecture and utility of the MobileGym platform. The writing is generally professional, with strong use of active voice and precise terminology in the system design sections. However, there are specific instances where sentence structure and clarity could be improved to ensure the reader does not stumble over grammatical disjointedness or ambiguous comparisons.

In the Introduction (Section 1), the list of limitations regarding everyday apps suffers from a lack of parallel structure. The phrase “Everyday apps are unreadable, unwritable, unforkable, and actions are irreversible” mixes adjectives describing the apps with a full clause describing actions. This creates a slight rhythmic break and potential confusion. A unified structure, such as “Everyday apps are unreadable, unwritable, and unforkable, and their actions are irreversible,” would improve flow.

In Section 5.2, the comparison of Sim-to-Real gains is slightly opaque. The text states the real-device pass rate rose “matching sim gain (33.9% $\square$ 76.7%, -42.8 pt).” Without explicitly restating that the 33.9% refers to the simulation baseline (which differs from the real-device baseline of 32.2%), a quick reader might conflate the two baselines. Clarifying the reference point for the percentages would eliminate this ambiguity.

Additionally, in Appendix A, the sentence regarding Hot-module replacement (HMR) contains a LaTeX artifact ($\$$) and a logical disconnect, suggesting HMR allows edits to take effect “in” synthetic entities. Since HMR affects the runtime code, not the data entities, the

phrasing should be adjusted to reflect that it facilitates the development process for these entities.

Overall, the paper is well-written, but these minor syntactic and clarity issues should be addressed to ensure the high-quality presentation matches the technical contribution.

Action items.

- [writing] In Section 1 (Introduction), the sentence ‘Everyday apps are unreadable, unwritable, unforkable, and actions are irreversible’ is grammatically disjointed. The first three adjectives lack a clear subject (implied ‘are’), while the fourth clause introduces a new subject (‘actions’). Rephrase for parallel structure, e.g., ‘Everyday apps are unreadable, unwritable, and unforkable, and their actions are irreversible.’
- [writing] In Section 5.2 (Sim-to-Real Transfer), the phrase ‘matching sim gain (33.9%–76.7%, –42.8 pt)’ is ambiguous. It is unclear if the 33.9% refers to the base model’s real-device performance or the simulation performance. Explicitly state ‘matching the simulation-side gain (33.9%–76.7% in sim vs. 32.2%–72.9% on real device)’ to prevent misinterpretation of the baseline.
- [writing] In Appendix A (System Implementation Details), the sentence ‘Hot-module replacement (HMR) allows code edits to take effect in

190Ksyntheticentities'containsaLaTeXformattingerror(

) and a semantic mismatch. HMR applies to the codebase, not the entities. Correct to ‘...allows code edits to take effect instantly, facilitating the management of 190K synthetic entities’ or similar.